

Abalone Age Prediction

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Introduction

Problem Statement

- **Objective:** Predict the age of abalones based on physical measurements.
- **Context:** The age of abalone, a marine mollusk, is a crucial factor for fisheries management and biological studies. Accurate age prediction can help in sustainable fisheries management, conservation efforts, and understanding the growth patterns of abalone populations.
- **Problem:** Given a set of physical attributes of abalones, such as Length, Diameter, Height, weights, and Sex, develop a predictive model to estimate the age of the abalone.

Application

Fisheries Management

- By understanding the age distribution of abalone populations through predictive models, fisheries can gain insights into the current state of the population. This information helps in identifying age classes that are over- or under-represented.
- **Regulation Establishment:**
 - With accurate age distribution data, fisheries can set more informed regulations.
 - **Size Limits:** Ensuring that abalones are not harvested before reaching a certain size, which often correlates with maturity.
 - **Age Limits:** Setting minimum and maximum age limits to protect both juvenile and older, reproductive individuals.
 - **Age-Based Quotas:** Implementing quotas based on age classes can help manage the harvest more effectively. For instance, limiting the number of abalones harvested from specific age groups can help maintain a balanced population structure.

Data

Dataset

Name	Type	Description	Role
Sex	Category	M, F, and I (infant)	Feature
Length	Continuous	Longest shell measurement	Feature
Diameter	Continuous	perpendicular to length	Feature
Height	Continuous	with meat in shell	Feature
Whole_weight	Continuous	whole abalone	Feature
Shucked_weight	Continuous	weight of meat	Feature
Viscera_weight	Continuous	gut weight (after bleeding)	Feature
Shell_weight	Continuous	after being dried	Feature
Rings	Integer	+1.5 gives the age in years	Target

Methodology

Methodology

Preprocessing

1. Encoding Categorical Data
2. Scaling Numeric Data with Standard Scaler

Training

1. Each training is done with GridSearch of 5
2. Neural Network is train with Epoch of 100

Models

Models

1. Ridge Regression
2. Decision Tree
3. Random Forest
4. XGBoost
5. Neural Network

1. Ridge Regression

- Ridge Regression is beneficial when dealing with multicollinearity and a large number of features.
- It helps to stabilize the estimates of regression coefficients by adding a penalty to their magnitude, thus improving model performance and interpretability in cases of correlated predictors.

2. Decision Tree

- Decision Trees provide an interpretable model that can handle both numerical and categorical data.
- They are capable of capturing non-linear relationships and interactions between features.
- Their visual representation makes them easy to understand and explain.

3. Random Forest

- Random Forest improves upon Decision Trees by combining multiple trees to reduce overfitting and improve prediction accuracy.
- The ensemble approach leverages the strength of individual trees while mitigating their weaknesses, making it robust and effective for a variety of tasks.

4. XGBoost

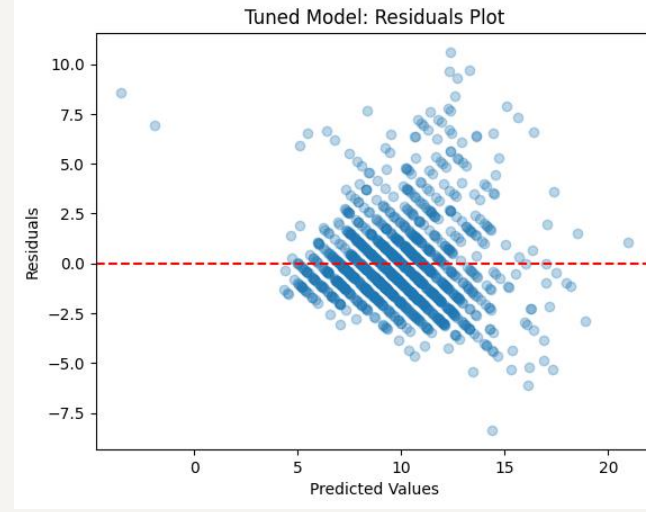
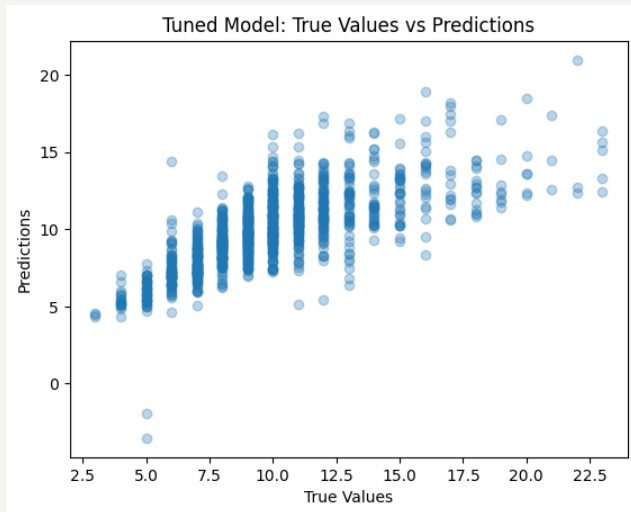
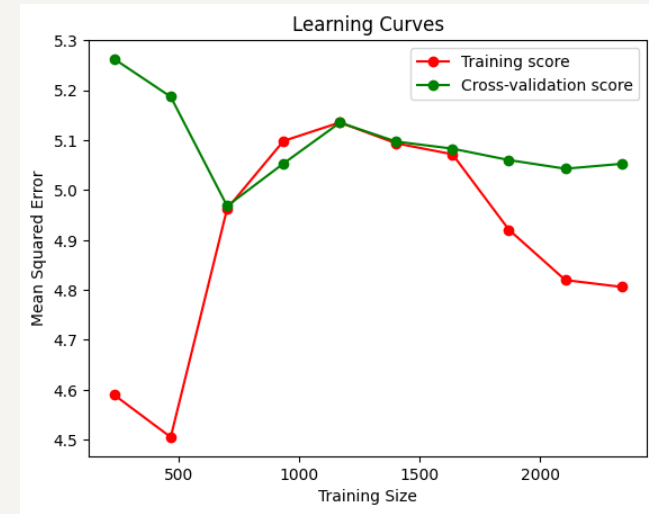
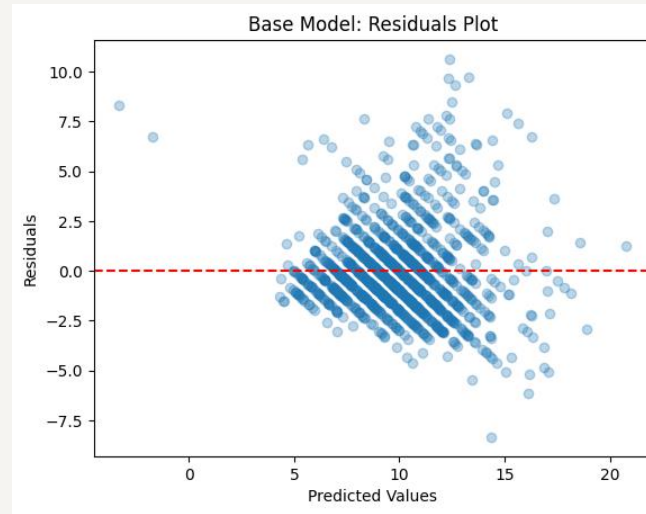
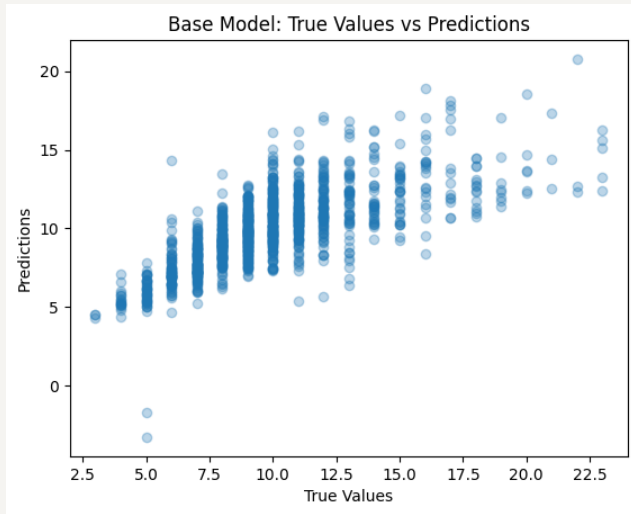
- XGBoost is known for its high performance in prediction accuracy and computational efficiency.
- Its gradient boosting framework allows it to handle complex data structures and improve model performance through iterative correction of errors.

5. Neural Network

- Neural Networks are highly flexible and powerful for modeling complex and non-linear relationships.
- They are particularly effective when dealing with large datasets and intricate patterns.
- The ability to adjust network architectures allows for customization based on specific problem requirements.

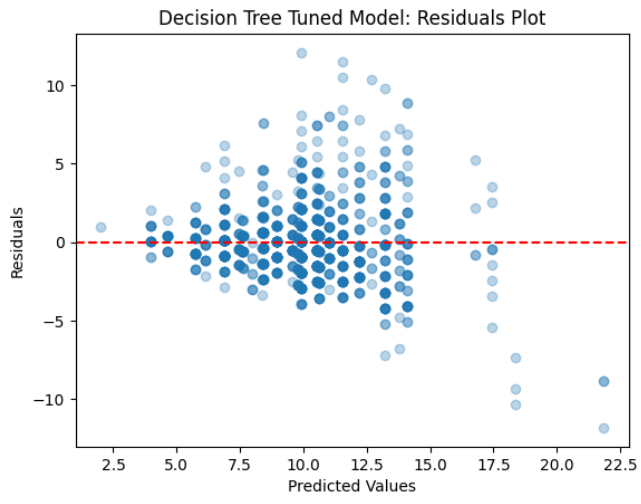
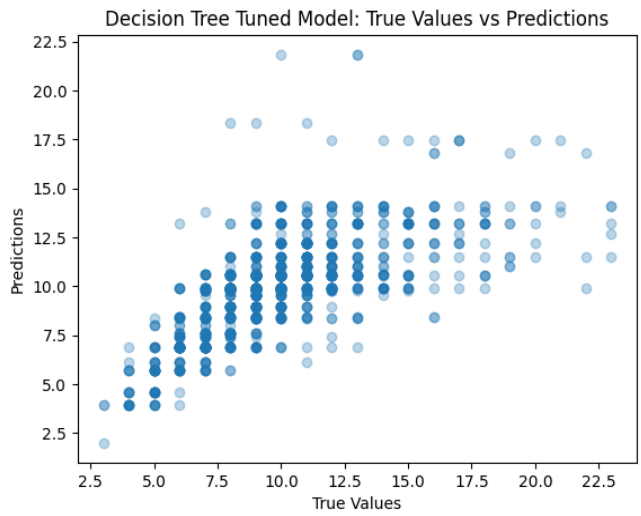
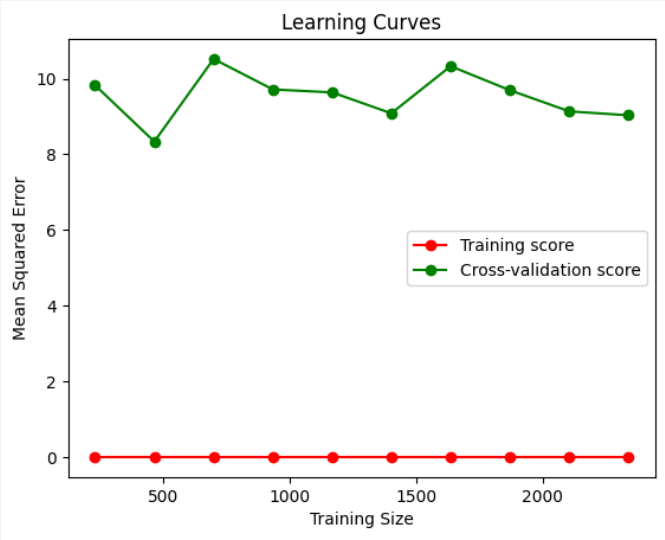
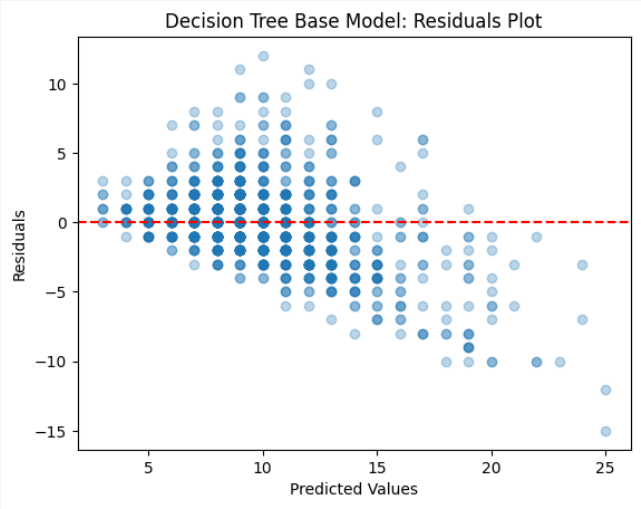
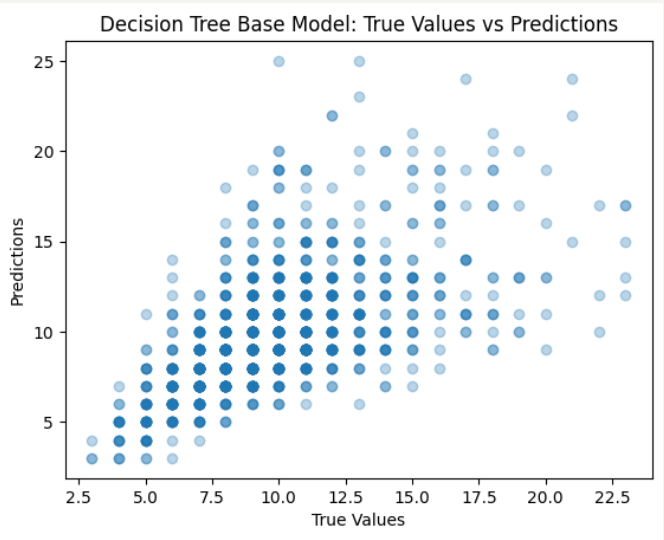
Solution and Evaluation

Ridge Regression



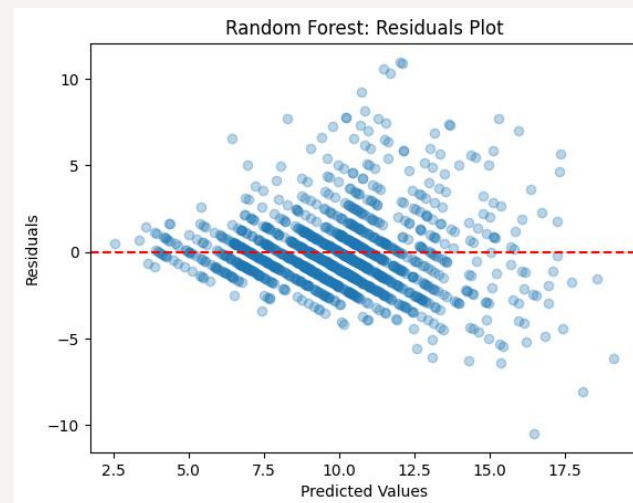
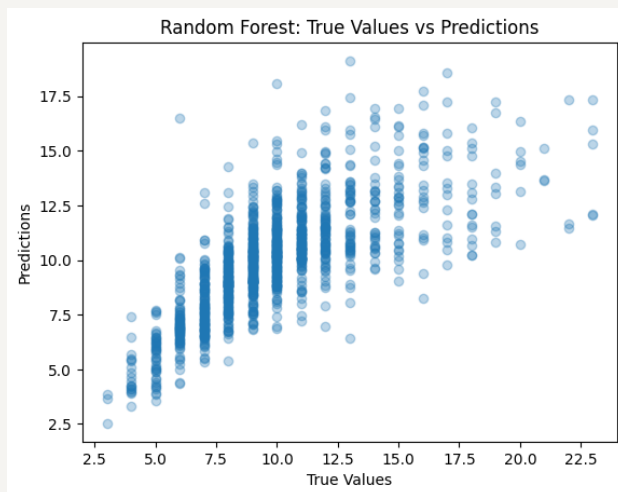
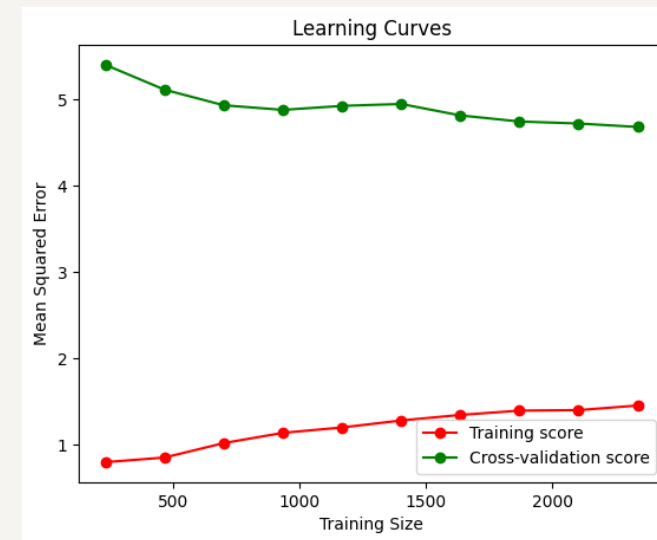
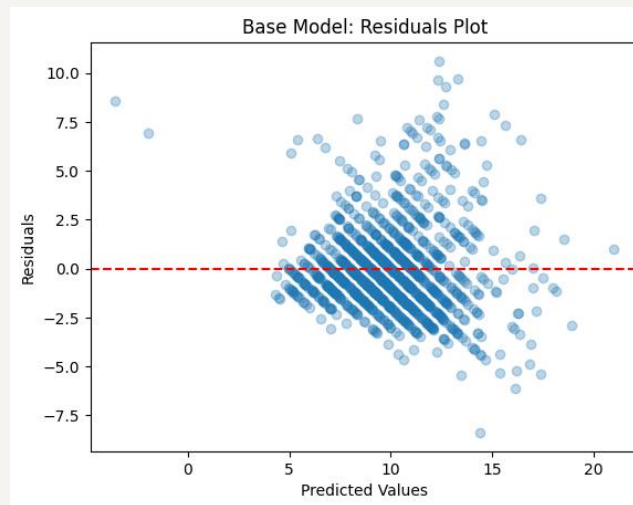
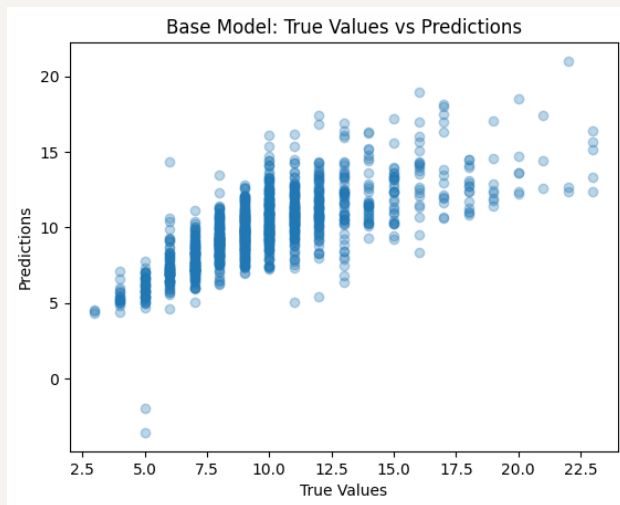
Model	MSE	RMSE	MAE	R-Squared	Adjust R-Squared
Ridge Regression (Base)	4.7757	2.1853	1.5829	0.5297	0.5259
Ridge Regression (Tuned)	4.7836	2.1872	1.5832	0.5289	0.5251

Decision Tree



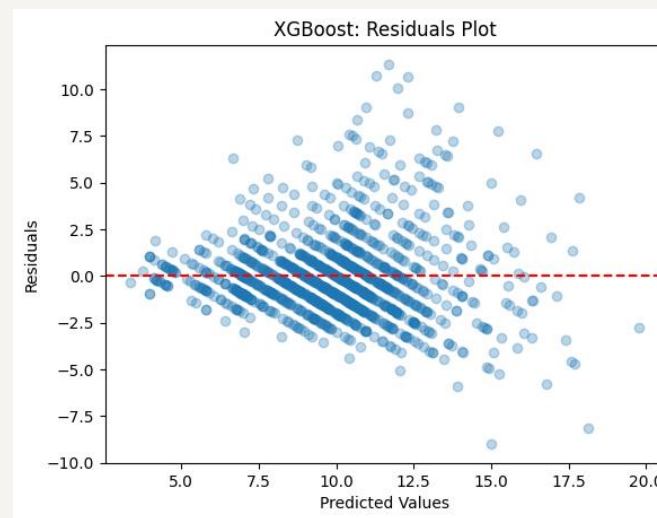
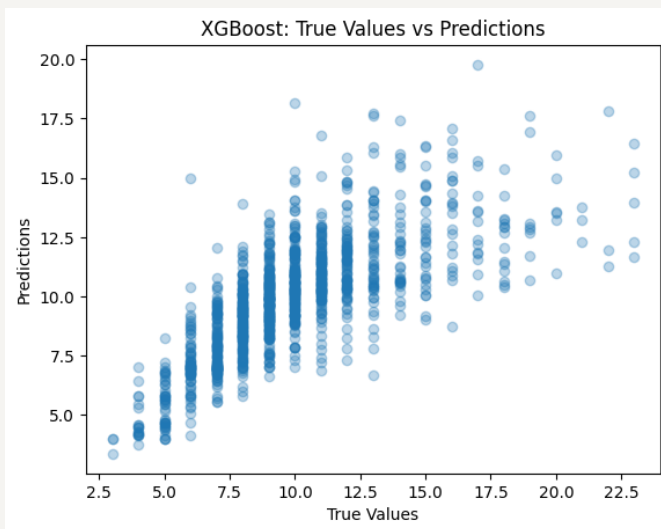
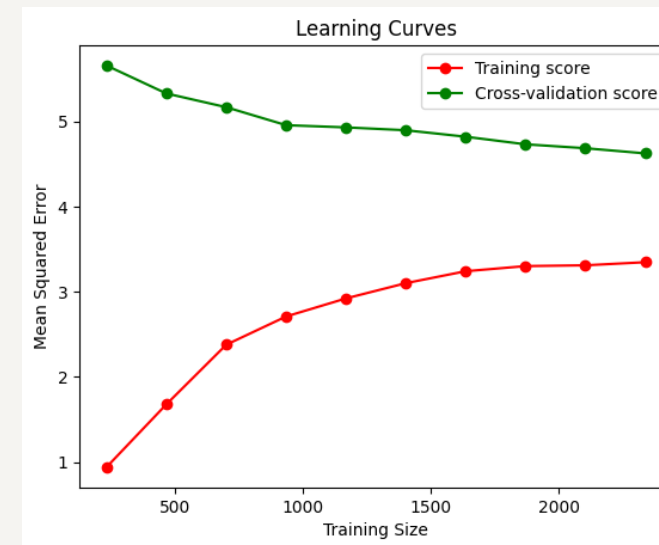
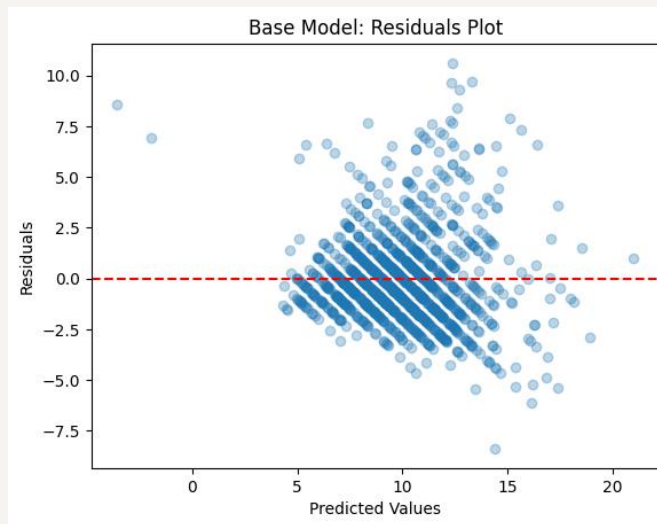
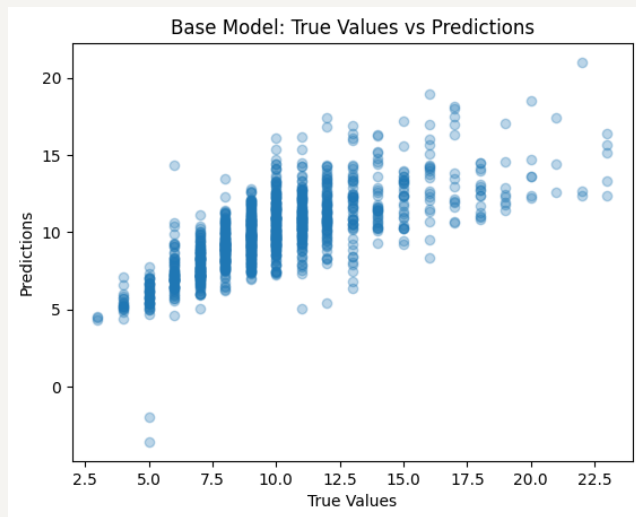
Model	MSE	RMSE	MAE	R-Squared	Adjust R-Squared
Decision Tree (Base)	8.3246	2.8852	2.0167	0.1802	0.1736
Decision Tree (Tuned)	5.4249	2.3291	1.6287	0.4658	0.4615

Random Forest



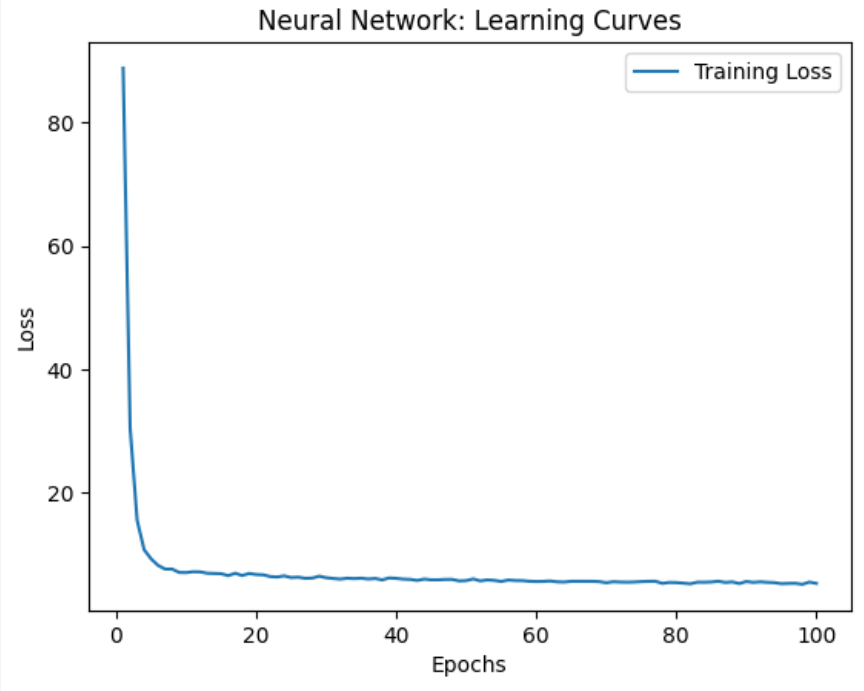
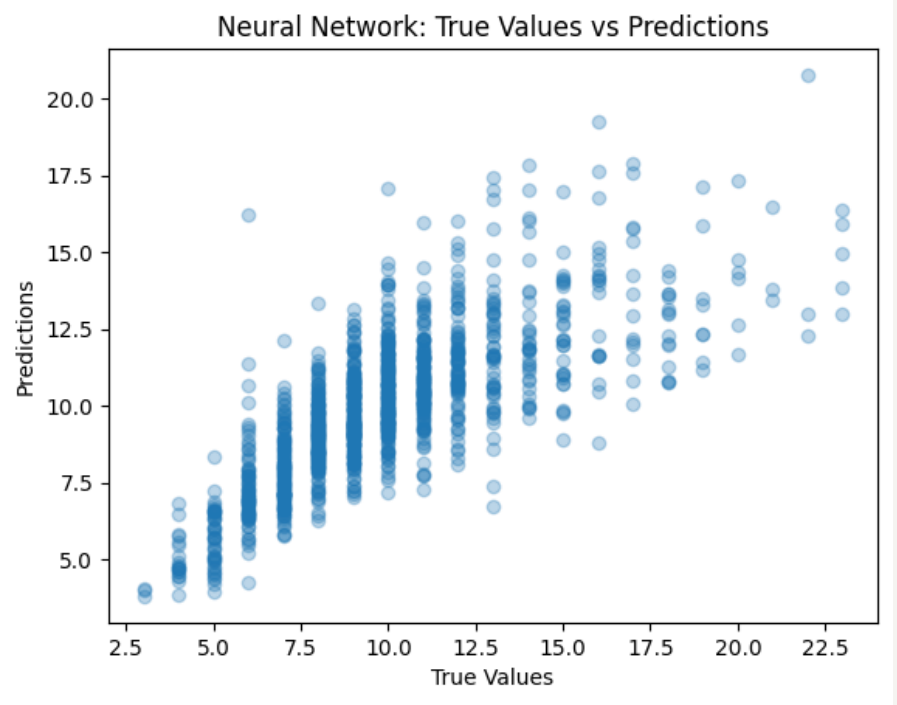
Model	MSE	RMSE	MAE	R-Squared	Adjust R-Squared
Random Forest (Base)	4.7848	2.1874	1.5832	0.5288	0.5250
Random Forest (Tuned)	4.7205	2.1727	1.5284	0.5351	0.5314

XGBoost



Model	MSE	RMSE	MAE	R-Squared	Adjust R-Squared
XGBoost (Base)	4.7848	2.1874	1.5832	0.5288	0.5250
XGBoost (Tuned)	4.6911	2.1659	1.5201	0.5380	0.5343

Neural Network



Model	MSE	RMSE	MAE	R-Squared	Adjust R-Squared
Neural Network	4.3507	2.0858	1.4860	0.5716	0.5681

Results

Model	MSE	RMSE	MAE	R-Squared	Adjust R-Squared
Ridge Regression (Base)	4.7757	2.1853	1.5829	0.5297	0.5259
Ridge Regression (Tuned)	4.7836	2.1872	1.5832	0.5289	0.5251
Decision Tree (Base)	8.3246	2.8852	2.0167	0.1802	0.1736
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XGBoost (Tuned)	4.6911	2.1659	1.5201	0.5380	0.5343
Neural Network	4.3507	2.0858	1.4860	0.5716	0.5681

The Neural Network model was the most effective for predicting abalone age, offering the highest accuracy but requiring more computational resources, while XGBoost provided a strong balance between performance and simplicity.

Q&A

References

- <https://www.ibm.com/topics/ridge-regression>
- <https://harshithaindumathi.medium.com/abalone-age-prediction-607ecda3fa3b>

Source

Source Code:

<https://colab.research.google.com/drive/1HJnbgoiaboPHu4qMaqK1mm7INJB0-O40?usp=sharing>